

PlayIQ Module 3: The Pre-Learn System

Your mission: make tomorrow's lesson feel familiar today

Have you ever walked into class, sat down, and within five minutes felt completely overwhelmed because the teacher was throwing out terms, diagrams, and formulas you've never seen before?

Your brain freezes. You scramble to copy down notes you don't understand. You leave class feeling behind before you even started.

Here is the secret top performers know: **they don't try to learn everything during the lecture.**

They use a **Pre-Learn System**.

Pre-learning is not doing tomorrow's homework in advance. It is not spending two hours reading a boring textbook chapter line-by-line.

Pre-learning is giving your brain a **rough map** before you enter the territory. When you already know the main landmarks, the five key words, and the one question to listen for, your brain stops panicking and starts connecting dots. You walk into class confident, calm, and two steps ahead.

By the end of this module, you will be able to:

- transform any complex, intimidating school topic into a clean, 5-part mental map in under 10 minutes;
- extract the "First Principle" (the core idea that makes all the details make sense);
- run the **Learn How You Learn Lab** to discover whether your brain prefers **Map-First**, **Example-First**, or **Question-First** entry;
- use **Non-Examples** to establish razor-sharp conceptual boundaries;
- build and test a 10-line Pre-Learn Brief;
- stage **Rule 3 ('pre-learn-sequence')** into your personal AI Tutor blueprint;
- defeat the **Pre-Learn Pioneer Boss Battle** to earn your Module 3 badge.

Estimated time: 45 to 60 minutes.

Part 0: Lightning Hook — Choose Your Arrival Style

Time: 60 seconds

Imagine you are dropped into a massive open-world video game or a theme park the size of a city. You have three ways to prepare before stepping through the gate:

- **Option A (The Atlas Guide):** You study a top-down aerial map showing the 5 major zones and how they connect. You don't know what every shop sells, but you know where the mountains, rivers, and castles are.
- **Option B (The Gameplay Trailer):** You watch a 30-second clip of an intense boss battle in action. Seeing real combat immediately shows you what the game feels like and how the mechanics work.
- **Option C (The Quest Log):** You are given one intriguing mystery: **"Why does the castle gate only open during an eclipse?"** You enter looking specifically for that one clue.

Which one makes your brain feel most ready to explore?

My choice: [Option A | Option B | Option C]

Why did that choice appeal to you?

There is no "wrong" choice.

- Option A represents **Map-First** learning (big-picture structure first).
- Option B represents **Example-First** learning (concrete action first).
- Option C represents **Question-First** learning (curiosity hook first).

In today's lab, you will run a live head-to-head trial with Orion to discover which entry method gives your brain the highest clarity and lowest cognitive overwhelm.

Part 1: Why Pre-Learning Destroys Classroom Overwhelm

Cognitive Load and Working Memory

Your brain's working memory can only hold about 4 to 7 new pieces of information at once. When a teacher introduces 15 new terms, 3 formulas, and a historical timeline all in 45 minutes, your working memory crashes like a computer trying to run 50 apps at once. Psychologists call this **cognitive overload**.

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WITHOUT PRE-LEARNING:

Teacher speaks → 15 new terms hit your brain at once → Buffer overflow → Overwhelm & zoned out

WITH PRE-LEARNING:

5-min Pre-Learn Map → 5 landmarks already saved in long-term memory → Teacher speaks → New details easily snap into place!

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The 5-Part Map

A high-speed Pre-Learn Map does not summarize every paragraph. It answers exactly five questions:

1. **The Headline:** What is the overarching topic in one sentence?
2. **The 5 Landmarks:** What are the 4 to 5 main parts or phases?
3. **The Power Vocabulary:** Which 3 to 5 critical terms will the teacher repeat most?
4. **The Anchor Example:** What is one real-world, everyday example that illustrates the concept?
5. **The Golden Question:** What is the single most interesting question to listen for in class?

Real-World Models Across Three Subjects

Look at how a 5-Part Map simplifies three notoriously difficult subjects:

Model 1: Biology — Cellular Respiration

- **Headline:** How living cells break down glucose (food) to create usable energy (ATP).
- **5 Landmarks:**
 1. Glycolysis (splitting sugar in the cytoplasm)
 2. Krebs Cycle (spinning carbon rings inside the mitochondria)
 3. Electron Transport Chain (the cellular power generator)
 4. ATP (the energy battery produced)
 5. Byproducts (carbon dioxide and water)
- **Power Vocabulary:** Glucose, ATP, Mitochondria, Aerobic, Byproduct.
- **Anchor Example:** Why your leg muscles burn during a sprint (when oxygen runs low, cells switch to lactic acid fermentation).
- **Golden Question:** "Where exactly inside the mitochondria is the majority of ATP created?"

Model 2: World History — Causes of World War I

- **Headline:** How decades of competing treaties, empires, and military buildups exploded into a global war after a single assassination.
- **5 Landmarks:**
 1. Militarism (arms race between navies and armies)

2. Alliances (secret defense pacts that dragged countries in like dominoes)
 3. Imperialism (fights over colonies in Africa and Asia)
 4. Nationalism (intense ethnic pride and desire for self-rule)
 5. The Spark (the assassination of Archduke Franz Ferdinand in Sarajevo)
- **Power Vocabulary:** Alliance, Mobilization, Deterrence, Sovereignty, Domino Effect.
 - **Anchor Example:** Two rival friend groups in a school hallway where a conflict between two people suddenly pulls 30 people into a fight because of pacts.
 - **Golden Question:** "Why did the alliances force countries to declare war even when their leaders didn't want to fight?"

Model 3: Algebra II — Logarithms

- **Headline:** The mathematical inverse of exponents—asking "what power do I need to reach this number?"
 - **5 Landmarks:**
1. Base (the number being multiplied)
 2. Exponent/Power (how many times it multiplies)
 3. Result/Argument (the final total)
 4. Common Log (base 10)
 5. Natural Log ($\ln$, base e)
- **Power Vocabulary:** Base, Exponent, Inverse, Argument, Logarithmic Scale.
 - **Anchor Example:** The Richter scale for earthquakes (a magnitude 7 earthquake isn't 1 point stronger than a 6; it's 10 times stronger).
 - **Golden Question:** "Why can you never take the logarithm of a negative number?"

Part 2: The Orion Map Protocol

Let's test generating a high-impact Pre-Learn Map using Orion (or your frontier AI workspace).

The Pre-Learn Prompt Template

Copy and adapt this exact prompt:

```
```text
```

Act as my PlayIQ learning coach. I am about to learn [INSERT UPCOMING TOPIC] in my [SUBJECT] class.

Do not give me a long lecture, full chapter summary, or practice problems yet.

Give me a high-speed 5-Part Pre-Learn Map:

1. One-sentence Headline (plain English, no jargon).
2. The 5 Major Landmarks (bullet points).
3. The 5 Essential Vocabulary Words with 6-word plain definitions.
4. One relatable, real-world Anchor Example.
5. One intriguing Golden Question I should listen for in class.

Label any uncertainty and ask what grade level I need.

```
```
```

Try It Live: Your Pre-Learn Map

Pick an actual topic from one of your classes this week (or an upcoming topic you dread):

My Topic: _____

Class/Subject: _____

Paste Orion's response and extract your map:

1. Headline:

2. 5 Landmarks:

- Landmark 1: _____
- Landmark 2: _____
- Landmark 3: _____
- Landmark 4: _____
- Landmark 5: _____

3. 5 Essential Vocabulary Words:

1. _____ : _____
2. _____ : _____
3. _____ : _____
4. _____ : _____
5. _____ : _____

4. Anchor Example:

5. My Golden Question for Class:

Part 3: Learn How You Learn Lab – The 3-Way Entry Trial

Now comes the scientific experiment. We want to discover how your brain naturally processes unfamiliar conceptual territory.

We will test three distinct pedagogical entry methods on a fresh, moderately challenging topic (e.g., *Game Theory & The Prisoner's Dilemma*, *Plate Tectonics & Subduction Zones*, or *The Law of Supply & Demand*).

Topic Chosen for Lab: _____

Round 1: Map-First Protocol

Open a clean chat thread with Orion and prompt:

> *"Give me a top-down, structural 5-part map of [TOPIC]. Show only the architecture and categories. Do not explain the mechanics or give real-world examples yet."*

Read the response once. Immediately record your initial metrics:

- Initial Clarity (1-5): ____ / 5
- Curiosity Triggered (1-5): ____ / 5
- Feeling of Overwhelm (1-5, where 1 is zero overwhelm, 5 is total panic): ____ / 5
- What mental image formed first? _____

Round 2: Example-First Protocol

Start a fresh chat thread with Orion and prompt:

> **"Do not explain the theory or give me a map of [TOPIC] yet. Instead, give me two vivid, real-world examples in action and one non-example. End by asking me what pattern connects them."**

Read the response once. Immediately record your metrics:

- Initial Clarity (1-5): ____ / 5
- Curiosity Triggered (1-5): ____ / 5
- Feeling of Overwhelm (1-5): ____ / 5
- What pattern did your brain notice? _____

Round 3: Question-First Protocol

Start a third fresh chat thread with Orion and prompt:

> **"Do not explain [TOPIC] yet. Present me with a real-world dilemma or counterintuitive paradox that forces me to make a decision. Ask me what I would do, then explain how [TOPIC] solves that paradox."**

Read the response once and answer the dilemma. Record your metrics:

- Initial Clarity (1-5): ____ / 5
- Curiosity Triggered (1-5): ____ / 5
- Feeling of Overwhelm (1-5): ____ / 5
- Did grappling with the question make you care more about the answer? [Yes | No]

Quantitative Comparison Scorecard

| Entry Method | Clarity (1-5) | Curiosity (1-5) | Overwhelm (1-5) | Best Subject Fit |
|-------------------------|---------------|-----------------|-----------------|------------------|
| Round 1: Map-First | ____ | ____ | ____ | _____ |
| Round 2: Example-First | ____ | ____ | ____ | _____ |
| Round 3: Question-First | ____ | ____ | ____ | _____ |

Self-Discovery Analysis

1. Which entry method lowered your feeling of overwhelm the fastest? Why?

2. Which entry method made you genuinely curious rather than just feeling like you were doing schoolwork?

3. Did you notice a difference based on subject? (e.g., Do you prefer Example-First for Math, but Map-First for History?)

Part 4: Non-Examples Reveal the Boundary

One of the biggest traps in learning is **false recognition**—thinking you understand a concept when you only recognize the label.

The fastest way to test true understanding is by finding the **Non-Example**.

- An **Example** shows what a concept is.
- A **Non-Example** looks almost identical on the surface, but violates one critical rule of the concept.

Seeing where a concept **stops** working is what builds true expertise.

Model: The Boundary Detective

| Concept | True Example | Non-Example (Close Call!) | Why It Fails the Rule (The Boundary) |
|------------------|---|---|---|
| Direct Democracy | Citizens in ancient Athens voting directly on a proposed law in the assembly. | Citizens in Canada voting for a Member of Parliament to represent them in the House of Commons. | Canada is a *Representative* Democracy; citizens elect representatives rather than voting directly on the laws themselves. |
| Chemical Change | An iron nail rusting in moist air to form iron oxide. | Ice melting into liquid water on a warm kitchen counter. | Melting ice is a *Physical* Change; the water molecules (H ₂ O) remain chemically identical. |
| Alliteration | "Peter Piper picked a peck of pickled peppers." | "City life is simple and serene." | Even though 'city', 'simple', and 'serene' start with an 's' sound, alliteration requires the repetition of initial consonant sounds across stressed syllables. |

Your Turn: Find the Boundary

Pick one key concept from your current classes.

Concept: _____

Ask Orion:

> **"For [CONCEPT], give me one clear real-world example and one sneaky non-example that looks like an example but isn't. Explain the exact boundary line where it fails."**

- **My Example:** _____
- **My Non-Example:** _____
- **The Boundary Rule:** _____
- **Verification Source Checked (Textbook / Class notes / .edu site):** _____

Part 5: The 10-Line Pre-Learn Brief

Before you walk into class tomorrow, you should be able to write your **10-Line Pre-Learn Brief** in five minutes flat.

If you get stuck on a line, write **[CHECK IN CLASS]** instead of making something up. That blank is not a failure—it's an arrow pointing to what you need to pay attention to!

```text

=====

10-LINE PRE-LEARN BRIEF

=====

1. UPCOMING TOPIC: \_\_\_\_\_
2. CORE FIRST PRINCIPLE: \_\_\_\_\_
3. LANDMARK 1: \_\_\_\_\_
4. LANDMARK 2: \_\_\_\_\_
5. LANDMARK 3: \_\_\_\_\_
6. KEY WORD & MEANING: \_\_\_\_\_
7. ANCHOR EXAMPLE: \_\_\_\_\_
8. CLOSE NON-EXAMPLE: \_\_\_\_\_
9. WHAT STILL CONFUSES ME: \_\_\_\_\_
10. MY GOLDEN QUESTION FOR CLASS: \_\_\_\_\_

=====

The 60-Second Memory Recall Test

Put a piece of paper over your 10-Line Brief.

Without peeking, answer out loud or in writing:

- 1. What is the core first principle?
- 2. What is your golden question for class?

Did you remember them? [ Yes | Partly | Blanked ]

If you blanked, that is totally fine! Look back at your brief for 10 seconds, cover it, and try once more. That small effort of retrieval locks the concept into your long-term memory.

Part 6: Learning Supercharger Blueprint Update — Staging Rule 3

In Module 0, you established your baseline and handshake. In Module 1, you staged Rule 1 ('explanation-start'). In Module 2, you staged Rule 2 ('voice-preservation' & 'session-length').

Now, you are staging Rule 3 ('pre-learn-sequence').

This rule dictates how your personal AI Tutor will introduce brand-new, complex material whenever you start a study session or ask about an unfamiliar topic.

Rule 3 Specification

Select the setting supported by your Lab evidence:

- [ ] 'pre-learn-sequence: map-first' (Always give me the structural 5-part map and categories before diving into examples or details).
- [ ] 'pre-learn-sequence: example-first' (Always give me two concrete real-world cases and a non-example before showing theoretical structure).
- [ ] 'pre-learn-sequence: question-first' (Always hook me with an intriguing puzzle, dilemma, or question about what I already suspect before explaining).

Why this setting matched your brain's processing:

Orion review condition: \*Retest this rule whenever switching between STEM (math/physics) and Humanities (literature/history).\*

Part 7: Knowledge File Record ('student\_profile.yaml')

Append this record into your PlayIQ Knowledge File data block. This YAML format will be ingested by your personal tutor in Module 9:

```
```yaml
module: 3

module_name: "The Pre-Learn System"

record_type: "experiment_result"

student_profile_update:

active_rules:

  • rule_id: "rule-3-pre-learn-sequence"

setting: "[map-first | example-first | question-first]"

rationale: "[Explain why this lowered your overwhelm]"

review_condition: "Retest across STEM vs Humanities"
```

```
lab_data:

topic_tested: "[Topic tested in Part 3]"

clarity_score: "[1-5]"

overwhelm_reduction_score: "[1-5]"

preferred_entry_mode: "[map-first | example-first | question-first]"

mastery_artifacts:

  • artifact_name: "Pre-Learn Mission Card"

status: "completed"

verified_by: "student_and_source_check"

...
```

Part 8: Proof Artifact — Pre-Learn Mission Card

Fill out your official mission card. You can keep this in your binder or phone as proof of preparation:

```
```markdown

|-----|
| PLAYIQ PRE-LEARN MISSION CARD |
|-----|

| Student Name / Code: _____ |
| Target Class & Date: _____ |
| Upcoming Subject & Topic: _____ |
| |
| 5-Part Map Status: [x] Complete [] In Progress |
| Anchor Example: _____ |
| Sneaky Non-Example: _____ |
| The Boundary Rule: _____ |
| |
| Staged Tutor Rule 3: `pre-learn-sequence = [YOUR SETTING]` |
| |
| MY GOLDEN QUESTION FOR TOMORROW'S CLASS: |
| " _____ " |
| |
| Teacher / Source Verification Signature or Date: _____ |
|-----|

```
```

Part 9: Mastery Challenge — The Pre-Learn Pioneer Boss Battle

To unlock Module 4, complete the **Pre-Learn Pioneer Boss Battle**:

Mission Briefing

Pick one real class you have tomorrow or later this week.

1. Run a 10-minute Pre-Learn session using your chosen sequence ('map-first', 'example-first', or 'question-first').
2. Write your 10-Line Pre-Learn Brief.
3. Formulate your Golden Question.
4. **The Live Trial:** Walk into that class. When the teacher begins teaching, listen actively for your 5 landmarks and your Golden Question. If appropriate, ask your Golden Question or write down the exact moment the teacher answers it!

Post-Mission Debrief (Complete After Class)

- Did hearing the topic in class feel less overwhelming than usual? [Yes | No | Neutral]
- When the teacher mentioned terms you had already seen in your brief, what did that feel like?

- Was your Golden Question answered by the teacher, or did you have to ask?

- Does your staged Rule 3 feel solid, or do you want to adjust it?

[Keep As Is | Retest Next Week]

Beta Tester Feedback — Beta Route Only

Complete this section only if your PlayIQ account is in the beta-testing track:

1. How clear was the difference between Map-First, Example-First, and Question-First? (1-5): ____ / 5
2. Did the Non-Example exercise help you catch any misunderstandings? _____
3. How long did it take you to complete your 10-Line Brief? _____ minutes
4. What was the most fun or surprising part of this module? _____
5. What would make this pre-learn mission even better? _____

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